Introduction:

1.1 Motivation

Aviophobia (Fear of Flying) is one of the most prevalent anxiety disorders in the modern world, significantly impacting the quality of life for millions of individuals. This anxiety hinders career advancement by preventing business travel, limits family vacations, and creates a profound sense of frustration and distress.

The currently accepted method for treating such phobias is Gradual Exposure Therapy. In recent years, Virtual Reality (VR) technology has become an effective tool, allowing patients to be exposed to flight scenarios in a controlled environment. However, existing VR solutions suffer from a fundamental limitation: the disconnect between the virtual experience and the patient's actual physiological state.

In most current systems, the therapist is forced to rely on the patient's subjective reporting (e.g., "On a scale of 1 to 10, how afraid are you?"). The therapist lacks real-time visibility into objective metrics—such as heart rate, perspiration (GSR), or blood pressure variability—which indicate an approaching anxiety attack. This lack of data can lead to over-exposure, potentially causing re-traumatization, or conversely, under-exposure that results in ineffective treatment.

The motivation for this project stems from the critical need to bridge this gap using technology. We aim to create a "smart" ecosystem that does not merely project scenes but "senses" the patient, analyzes their medical status in real-time, and provides the therapist with Data-Driven Decision Making tools. This transformation aims to make therapy safer, highly personalized, and significantly more effective.

1.2 Project Description

This project focuses on the development of a management and control system for treating flight phobia, integrating IoT (Internet of Things) and AI (Artificial Intelligence) technologies within a VR environment.

The system interconnects three primary components:

1. The VR System: Runs simulations of various flight stages (Lobby, Takeoff, Cruising, Landing).
2. Wearable IoT Component (Samsung Watch): Collects physiological metrics in real-time (Heart Rate, Blood Pressure, HRV, GSR) directly from the patient.
3. The Management System (Web Application): Serves the Therapist and the Administrator (Owner).

How the System Works:

During the therapy session, data from the smartwatch is streamed continuously to the server. A Machine Learning (ML) engine analyzes this data, calculates a weighted Stress Index, and presents a precise status overview to the therapist via a dedicated Dashboard.

Beyond monitoring, the system includes an intelligent safety layer: Predictive Analysis algorithms identify patterns of escalating distress and alert the therapist before the patient reaches a state of panic. Furthermore, the AI can automatically recommend whether the patient is physiologically fit to proceed to the next stage of the flight simulation (e.g., moving from Takeoff to Landing).

At the conclusion of each session, the system generates detailed reports that allow for tracking the patient's progress over time, thereby transforming psychological treatment into a measurable, precise, and evidence-based practice.